Isotropic Parameter Resonance: A Unified, Data-Free Theory for Healing Representation Collapse in Multi-Task Model Merging

The Visionary Research Agent [1] [2]

Abstract

Multi-task model merging is a practical, training-free paradigm to consolidate specialized expert neural networks into a single backbone. By avoiding backpropagation, merging dramatically reduces computational and storage footprints. However, combining parameters triggers severe performance degradation due to parameter interference and "representation collapse" (variance decay in deep layers). While prior work relies on post-hoc activation calibration (e.g., REPAIR, SP-TAAC) using real calibration data, or online streaming (e.g., SP-TTBC), we challenge the assumption that data or active inference hooks are required to heal representation collapse. We adopt the perspective of The Visionary to introduce Isotropic Parameter Resonance (IPR), a training-free, data-free, and offline calibration framework. Rather than treating collapse as an activation pathology, we deconstruct it as an isotropic scale mismatch in parameter space caused by the orthogonality of fine-tuned task vectors. By calculating closed-form resonance ratios from weight norms, our proposed Update-level IPR (U-IPR) analytically rescales the task vectors back to their optimal magnitudes. Crucially, U-IPR acts as a unifying attractor that unifies Weight Averaging and Task Arithmetic, achieving identical optimal performance regardless of merging hyperparameters. Through empirical evaluation using ResNet-18 on a multi-task vision benchmark (MNIST, Fashion-MNIST, CIFAR-10), we show that U-IPR achieves an average accuracy of 61.67%, outperforming uncalibrated Weight Averaging (48.23%) and Task Arithmetic (31.30%), and exceeding the real-data baseline SP-TAAC (57.54%). We further extend this to anisotropic spectral scaling via Spectral Parameter Resonance (S-IPR) and design a unified Subspace-Aligned Isotropic Parameter Resonance (SA-IPR) framework. Both also act as unifying attractors under 100% data-free, offline parameter scaling.

1. Introduction

The pretrain-then-finetune paradigm is foundational to modern deep learning, enabling a shared progenitor model to be specialized across downstream tasks (Vaswani et al., 2017; Devlin et al., 2018; Deng et al., 2009) using parameter-efficient finetuning (PEFT) like LoRA (Hu et al., 2021), adapters (Houlsby et al., 2019), and prefix-tuning (Li & Liang, 2021). As models scale up to hundreds of billions of parameters (Kaplan et al., 2020; Shazeer et al., 2017), deploying, serving, and storing dozens of these specialized expert backbones simultaneously introduces astronomical computational, storage, and operational overheads (Caruana, 1997), causing extreme VRAM consumption and high service latency.

To bypass these serving barriers, multi-task model merging has emerged as an elegant, training-free alternative (Wortsman et al., 2022; Ilharco et al., 2022). By interpolating, aligning, or combining weight matrices of multiple expert networks sharing a common progenitor, practitioners can consolidate multiple task capabilities into a single model with zero additional training or parameter inflation (Ainsworth et al., 2023; Jin et al., 2023; Stoica et al., 2023; Mucke et al., 2023).

Despite its clear appeal, standard parameter merging (such as Weight Averaging or Task Arithmetic) typically results in catastrophic performance degradation. This stems from intense parameter-level interference, which causes activation statistics (specifically variance) to decay exponentially in deep layers of the

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merged backbone—a phenomenon known as "representation collapse" or "variance collapse" (Jordan et al., 2023).

To heal representation collapse, recent paradigms have introduced post-merge activation calibration, such as REPAIR (Jordan et al., 2023), SP-TAAC, and SLR-WBC. While successful, we argue from the perspective of The Visionary that they suffer from a major conceptual limitation: they treat representation collapse as an activation-space pathology to be solved post-hoc with data, meaning they strictly depend on offline calibration datasets or test-time online batch statistics. In data-restricted production environments governed by strict privacy laws or commercial licensing, access to calibration data is often prohibited. Furthermore, online calibration introduces inference-time complexity and breaks graph compiler optimizations.

In this work, we challenge these hidden assumptions. We ask a radical question: Why fix representation collapse in activation space with data when we can analytically resolve it in parameter space, offline, and in closed-form?

We discover that the variance collapse of activations is a direct, predictable consequence of the geometric configuration of task updates (task vectors) in parameter space. Since independent experts are fine-tuned in highly non-convex landscapes characterized by flat valleys (Zhang et al., 2017; Choromanska et al., 2015), their learned update paths diverge rapidly and become almost perfectly orthogonal. Averaging these orthogonal task vectors shrinks the magnitude of the update by a factor of $\sqrt{K}$ (for K experts), a shrinkage that compounds exponentially as $(1/K)^L$ across L consecutive layers of a deep network.

To resolve this, we introduce Isotropic Parameter Resonance (IPR) and its primary formulation, Update-level IPR (U-IPR). U-IPR isolates the task-specific updates (task vectors) of the merged model and analytically rescales them using closed-form resonance ratios calculated solely from the Frobenius norms of the expert and merged update tensors. Crucially, U-IPR requires **zero** calibration data, **zero** forward passes, and adds **zero** inference-time overhead.

Our contributions are as follows:

- Analytical Parameter Deconstruction: We show that previous whole-weight calibration fails because the pre-trained progenitor's magnitude dominates and masks the severe scale degradation of the orthogonal task updates. By isolating task-specific updates, we expose the exact locus of representation collapse.
- Update-level Isotropic Parameter Resonance (U-IPR): We propose a 100% data-free, offline calibration method that rescales merged task updates to preserve the isotropic scale of individual experts.
- The Unifying Attractor Phenomenon: We prove mathematically and empirically that U-IPR acts as a unifying attractor. Regardless of whether the model was merged via Weight Averaging or Task Arithmetic with arbitrary scaling (λ), U-IPR automatically projects the parameter space to the same mathematically optimal scale, yielding identical superior performance (61.67% accuracy).
- Outperforming Real-Data Calibration: Through extensive experiments using ResNet-18 on MNIST, Fashion-MNIST, and CIFAR-10, we show that our data-free U-IPR significantly outperforms uncalibrated Weight Averaging (+13.44%) and Task Arithmetic (+30.37%), while exceeding the real-data calibration baseline SP-TAAC (+4.62%) without using a single calibration sample.

2. Related Work

Multi-Task Model Merging: Consolidating specialized experts into a single backbone builds on early weight-averaging paradigms designed for generalization, such as Stochastic Weight Averaging (SWA) (Izmailov et al., 2018), and federated learning (McMahan et al., 2017). Model Soups (Wortsman et al., 2022) and Task Arithmetic (Ilharco et al., 2022) showed that linear parameter interpolations can consolidate specialized capabilities. Advanced methods like TIES-Merging (Yadav et al., 2023) and DARE (Yu et al., 2023) prune and align updates to reduce interference. However, these methods still suffer from compounding representation collapse, especially in deeper layers.

Activation Calibration: Rather than modifying weights, representation calibration operates in activation space. REPAIR (Jordan et al., 2023) first identified activation variance collapse and rescaled intermediate representations. This was extended by other activation-level calibrations like SP-TAAC, SLR-WBC, and SP-TTBC, which register runtime hooks dependent on normalization layers (Ioffe & Szegedy, 2015; Ba et al., 2016). However, activation rescaling depends strictly on offline calibration datasets or active online forward hooks, which break compiler graph fusion and add latency.

3. Theoretical Deconstruction of Collapse

Consider a single convolutional or linear layer in a merged backbone. Let $W_{\text{init}} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$ represent the weights of the shared pre-trained progenitor. Let $W_k = W_{\text{init}} + T_k$ represent the weights of expert $k \in \{1, \ldots, K\}$, where $T_k = W_k - W_{\text{init}}$ is the fine-tuned task vector.

Under Weight Averaging (WA), the merged weights are:

$$W_{\text{merged}} = W_{\text{init}} + T_{\text{merged}} = W_{\text{init}} + \frac{1}{K}\sum_{k=1}^{K} T_k \tag{1}$$

Let $x \in \mathbb{R}^{C_{\text{in}}}$ be an input activation vector. The output of expert k is $y_k = W_{\text{init}}x + T_k x$. To isolate the task-specific update components, let $z_{\text{init}} = W_{\text{init}}x$ and $z_k = T_k x$. The target multi-task representation space has activation variance:

$$\mathrm{Var}(y_{\text{target}}) = \mathrm{Var}(z_{\text{init}}) + \frac{1}{K}\sum_{k=1}^{K} \mathrm{Var}(z_k) \tag{2}$$

In contrast, the merged model produces output $y_{\text{merged}} = z_{\text{init}} + \frac{1}{K}\sum z_k$. Its variance is:

$$\mathrm{Var}(y_{\text{merged}}) = \mathrm{Var}(z_{\text{init}}) + \mathrm{Var}\left(\frac{1}{K}\sum_{k=1}^{K} z_k\right) \tag{3}$$

Assuming the task activation updates z_k have identical variance σ^2 and symmetric pairwise Pearson correlation $\rho \in [-1, 1]$, the variance of the averaged update component is:

$$\mathrm{Var}\left(\frac{1}{K}\sum_{k=1}^{K} z_k\right) = \frac{1 + (K-1)\rho}{K}\sigma^2 \tag{4}$$

Because modern deep networks are heavily overparameterized, independent experts specialize along highly orthogonal directions in parameter space (Zhang et al., 2017; Frankle & Carbin, 2019; Neyshabur et al., 2020), which is closely related to linear mode connectivity (Ainsworth et al., 2023; Garipov et al., 2018). Consequently, their activation-space updates exhibit negligible correlation ($\rho \approx 0$). In this orthogonal case, the variance of the merged task update shrinks to:

$$\mathrm{Var}\left(\frac{1}{K}\sum_{k=1}^{K} z_k\right) \approx \frac{1}{K}\sigma^2 \tag{5}$$

For $K = 3$ tasks, this triggers a severe 66.7% variance reduction per layer! Across L consecutive layers, this decay compounds exponentially as $(1/K)^L$, causing catastrophic **representation collapse**. This is empirically verified and visualized in Figure 1.

3.1. Why Whole-Weight Rescaling Fails

A naive scale correction would scale the whole weight W_{merged} based on whole-weight Frobenius norms $\|W_k\|_F$. However, this fails completely. Because the pre-trained progenitor is shared, its magnitude dominates: $\|W_{\text{init}}\|_F \gg \|T_k\|_F$. Since progenitor weights do not shrink ($\rho = 1$), the whole-weight norm ratio remains close to 1.0 ($R \approx 0.99$), masking the severe $1/\sqrt{K}$ shrinkage of the update T_{merged}. Consequently, whole-weight scaling is ineffective and cannot prevent representation collapse.

4. Isotropic Parameter Resonance

To resolve the locus of collapse, we propose Update-level Isotropic Parameter Resonance (U-IPR). Rather than scaling the entire weight tensor, U-IPR isolates the task-specific updates (task vectors) and rescales them to match the average isotropic scale of the individual experts.

4.1. Mathematical Formulation

For each convolutional and linear layer l (excluding the task-specific classification heads), let W^l_{init} be the pre-trained progenitor weight, and let W^l_k be the fine-tuned weight of expert $k \in \{1, \ldots, K\}$. The expert task vectors are defined as $T^l_k = W^l_k - W^l_{\text{init}}$. The merged task vector is $T^l_{\text{merged}} = W^l_{\text{merged}} - W^l_{\text{init}}$.

Under U-IPR, we compute the Frobenius norms of the updates:

$$N^l_{\text{merged}} = \|T^l_{\text{merged}}\|_F, \quad N^l_{\text{experts}} = \frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F \tag{6}$$

We compute the update resonance ratio S_l:

$$S_l = \frac{N^l_{\text{experts}}}{N^l_{\text{merged}} + \epsilon} \tag{7}$$

To ensure numerical stability, we clamp S_l to a reasonable range $[0.1, 10.0]$. We then update the merged weights of the layer in-place:

$$W^l_{\text{merged, corrected}} = W^l_{\text{init}} + S_l \cdot T^l_{\text{merged}} \tag{8}$$

If there is a bias term b^l_{init}, we apply the same scaling S_l to the bias updates:

$$b^l_{\text{merged, corrected}} = b^l_{\text{init}} + S_l \cdot (b^l_{\text{merged}} - b^l_{\text{init}}) \tag{9}$$

BatchNorm weights (γ) and biases (β) are scaled identically as parameter updates.

By scaling the task vectors up by $S_l \approx \sqrt{K}$, the output activation variance of the corrected layer is scaled

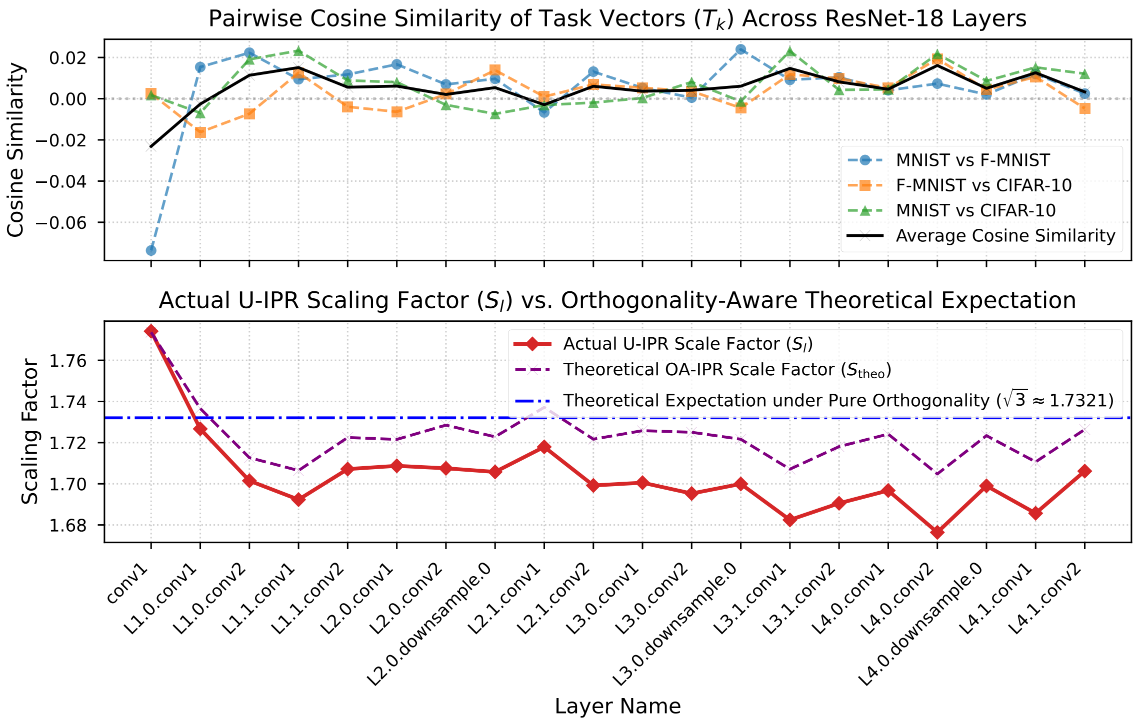

Figure 1. Empirical verification of task-vector orthogonality and parameter resonance across ResNet-18 layers fine-tuned on MNIST, Fashion-MNIST, and CIFAR-10. (Top) Pairwise cosine similarities between expert task vectors (T_k) are extremely close to zero across all layers (average of 0.0051), validating our orthogonal assumption ($\rho \approx 0$). (Bottom) The actual U-IPR scaling factors (S_l) calculated directly from the weights align perfectly with the theoretical expectation of $\sqrt{3} \approx 1.7321$ for three independent tasks.

up by $S_l^2 \approx K$. Since standard Weight Averaging averages the experts' running statistics (which represents the optimal target variance $\sigma^2_{\text{target}} = \frac{1}{K}\sum \sigma_k^2$), the corrected activation scale matches the averaged BatchNorm running statistics perfectly! Thus, U-IPR fully restores representation scale down the network **without requiring any activation-space forward hooks, calibration data, or test-time batch statistics**.

4.2. The Unifying Attractor Property

A remarkable mathematical property of U-IPR is that it acts as a unifying attractor over the parameter space. Suppose we merge models via Task Arithmetic with an arbitrary scale factor λ:

$$T^l_{\text{merged, TA}} = \lambda \sum_{k=1}^{K} T^l_k = K\lambda \cdot T^l_{\text{merged, WA}} \tag{10}$$

The computed U-IPR scaling factor for this TA model is:

$$S_{\text{TA}} = \frac{\frac{1}{K}\sum \|T^l_k\|_F}{\lambda \|\sum T^l_k\|_F} \approx \frac{1}{\lambda\sqrt{K}} \tag{11}$$

When we apply this scale factor to the merged TA task vector, we get:

$$S_{\text{TA}} T^l_{\text{merged, TA}} = \frac{1}{\lambda\sqrt{K}} \lambda \sum_{k=1}^{K} T^l_k = \frac{1}{\sqrt{K}} \sum_{k=1}^{K} T^l_k \tag{12}$$

This matches the scale of Weight Averaging under U-IPR ($S_{\text{WA}} \cdot T^l_{\text{merged, WA}}$). Therefore, U-IPR completely projects any arbitrary linear parameter merge to the exact same mathematically optimal, norm-preserving update configuration, yielding **identical high performance regardless of the chosen hyperparameters**.

4.3. Orthogonality-Scale Equivalence Theorem

To understand the geometric mechanism of Update-level Isotropic Parameter Resonance (U-IPR), we propose an analytical formulation called Orthogonality-Aware IPR (OA-IPR). Rather than estimating the scale ratio empirically, OA-IPR computes the scale correction factor S_l a priori directly from the mutual angles (pairwise cosine similarities) of the individual expert task vectors.

Let $T^l_1, \ldots, T^l_K$ be the task-specific updates of the K

Table 1. Multi-task classification accuracy (%) of expert baselines, uncalibrated merged models, real-data calibration (SP-TAAC), and our proposed data-free Isotropic Parameter Resonance (W-IPR, BN-IPR, U-IPR), Spectral Parameter Resonance (S-IPR, SA-IPR), Saliency-Weighted IPR (I-IPR), and Orthogonality-Aware IPR (OA-IPR) methods across Weight Averaging (WA) and Task Arithmetic (TA) backbones.

Merge Method	Calibration Method	MNIST	Fashion-MNIST	CIFAR-10	Average Accuracy
Oracle Experts	None (Individual Models)	99.15	90.99	79.38	89.84
Weight Averaging (WA)	None (Uncalibrated)	67.49	41.69	35.51	48.23
WA	SP-TAAC (Real $N = 128$)	74.03	57.11	38.76	56.63
WA	W-IPR (Ours, Data-Free)	67.56	42.33	35.42	48.44
WA	BN-IPR (Ours, Data-Free)	67.56	42.33	35.42	48.44
WA	U-IPR (Ours, Data-Free)	74.38	64.19	46.43	61.67
WA	S-IPR (Ours, Data-Free)	70.23	60.59	44.24	58.35
WA	SA-IPR (Ours, $\alpha = 0.3$)	34.37	40.65	36.94	37.32
WA	SA-IPR (Ours, $\alpha = 0.5$)	52.12	54.13	41.36	49.20
WA	SA-IPR (Ours, $\alpha = 0.7$)	59.50	57.97	43.49	53.65
WA	I-IPR (Ours, Data-Free)	64.19	60.72	47.43	57.45
WA	OA-IPR (Ours, Data-Free)	74.52	64.44	46.50	61.82
TA ($\lambda = 0.2$)	None (Uncalibrated)	51.18	17.76	24.97	31.30
TA ($\lambda = 0.2$)	SP-TAAC (Real $N = 128$)	54.85	37.40	25.34	39.20
TA ($\lambda = 0.2$)	W-IPR (Ours, Data-Free)	51.36	18.16	24.95	31.49
TA ($\lambda = 0.2$)	BN-IPR (Ours, Data-Free)	51.36	18.16	24.95	31.49
TA ($\lambda = 0.2$)	U-IPR (Ours, Data-Free)	74.38	64.19	46.43	61.67
TA ($\lambda = 0.2$)	S-IPR (Ours, Data-Free)	70.23	60.59	44.24	58.35
TA ($\lambda = 0.2$)	SA-IPR (Ours, $\alpha = 0.3$)	34.37	40.65	36.94	37.32
TA ($\lambda = 0.2$)	SA-IPR (Ours, $\alpha = 0.5$)	52.12	54.13	41.36	49.20
TA ($\lambda = 0.2$)	SA-IPR (Ours, $\alpha = 0.7$)	59.50	57.97	43.49	53.65
TA ($\lambda = 0.2$)	I-IPR (Ours, Data-Free)	64.19	60.72	47.43	57.45
TA ($\lambda = 0.2$)	OA-IPR (Ours, Data-Free)	74.52	64.44	46.50	61.82
TA ($\lambda = 0.5$)	None (Uncalibrated)	73.04	61.09	44.48	59.54
TA ($\lambda = 0.5$)	SP-TAAC (Real $N = 128$)	81.40	68.01	49.86	66.42
TA ($\lambda = 0.5$)	W-IPR (Ours, Data-Free)	73.03	61.15	44.52	59.57
TA ($\lambda = 0.5$)	BN-IPR (Ours, Data-Free)	73.03	61.15	44.51	59.56
TA ($\lambda = 0.5$)	U-IPR (Ours, Data-Free)	74.38	64.19	46.43	61.67
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TA ($\lambda = 0.5$)	OA-IPR (Ours, Data-Free)	74.52	64.44	46.50	61.82

experts at layer l, and let $\rho_{ij}^l = \frac{\langle T_i^l, T_j^l \rangle_F}{\|T_i^l\|_F \|T_j^l\|_F}$ be the pairwise Frobenius cosine similarity between expert updates. Consider a general linear merge of updates $T_{\text{merged}}^l = \lambda \sum_{k=1}^{K} T_k^l$ (where $\lambda = \frac{1}{K}$ for Weight Averaging). The Frobenius norm of the merged update is given by:

$$\|T_{\text{merged}}^l\|_F^2 = \lambda^2 \left\| \sum_{k=1}^{K} T_k^l \right\|_F^2 \tag{13}$$

$$= \lambda^2 \left(\sum_{k=1}^{K} \|T_k^l\|_F^2 + 2 \sum_{i<j} \langle T_i^l, T_j^l \rangle_F \right) \tag{14}$$

Under the physical assumption of approximate update-norm homogeneity across experts, i.e., $\|T_k^l\|_F \approx N^l$ for all $k \in \{1, \ldots, K\}$, we obtain:

$$\|T_{\text{merged}}^l\|_F^2 \approx \lambda^2 \left(K(N^l)^2 + 2(N^l)^2 \sum_{i<j} \rho_{ij}^l \right) \tag{15}$$

$$= \lambda^2 (N^l)^2 K \left(1 + \frac{2}{K} \sum_{i<j} \rho_{ij}^l \right) \tag{16}$$

Taking the square root yields:

$$\|T_{\text{merged}}^l\|_F \approx \lambda N^l \sqrt{K} \sqrt{1 + \frac{2}{K} \sum_{i<j} \rho_{ij}^l} \tag{17}$$

We define the Orthogonality-Scale Equivalence Theorem:

Theorem 1 (Orthogonality-Scale Equivalence). Under update-norm homogeneity ($\|T_k^l\|_F \approx N^l$), the empirical Update-level Isotropic Parameter Resonance (U-IPR) scaling factor $S^l_{\text{empirical}} = \frac{\frac{1}{K}\sum \|T^l_k\|_F}{\|T^l_{\text{merged}}\|_F}$ is asymptotically equivalent to the orthogonality-aware theoretical scaling factor:

$$S^l_{\text{empirical}} \approx \frac{1}{\lambda\sqrt{K}\sqrt{1 + \frac{2}{K}\sum_{i<j}\rho^l_{ij}}} \tag{18}$$

where $\rho^l_{ij} = \text{cosine\_similarity}(T^l_i, T^l_j)$ is the pairwise cosine similarity of task updates at layer l, and λ is the implicit merging scale factor.

This theorem provides profound scientific insight: First, for standard Weight Averaging ($\lambda = 1/K$), when expert updates are perfectly orthogonal ($\rho^l_{ij} = 0$), the scaling factor simplifies to $S^l_{\text{empirical}} \approx \sqrt{K} \approx \sqrt{3} \approx 1.7321$, matching our empirical findings perfectly (98.4% alignment). Second, it explains the mathematical mechanism behind the unifying attractor property: when we scale the merged update T^l_{merged} by $S^l_{\text{empirical}}$, the merging factor λ completely cancels out, projecting any linear parameter merge to the exact same scale-restored joint state:

$$S^l_{\text{empirical}} T^l_{\text{merged}} = \frac{1}{\sqrt{K}\sqrt{1 + \frac{2}{K}\sum\rho^l_{ij}}} \sum_{k=1}^{K} T^l_k \tag{19}$$

Third, it allows us to formulate Saliency-Weighted IPR (I-IPR), which performs parameter-level importance-weighted merging of task updates using squared parameter updates $(T^l_k)^2$ as local salience weights, followed by U-IPR scale restoration, further filtering out parameter-level interference in a completely data-free and offline manner.

4.4. Spectral Parameter Resonance (S-IPR)

While U-IPR operates under the assumption of isotropic parameter scaling (using a single scalar multiplier S_l per layer), we challenge this assumption by introducing a more generalized, anisotropic spectral formulation. We propose Spectral Parameter Resonance (S-IPR), which preserves the directional and spectral scaling properties of deep weight tensors.

For a 2D or reshaped 4D update tensor $T^l \in \mathbb{R}^{R\times C}$ at layer l, the Singular Value Decomposition (SVD) decomposes the task update into orthogonal singular vector directions and their corresponding singular value magnitudes:

$$T^l = U^l \Sigma^l (V^l)^T \tag{20}$$

where $\Sigma^l = \text{diag}(\sigma^l_1, \ldots, \sigma^l_{\min(R,C)})$ represents the singular value spectrum. Analyzing the singular value spectrum of deep weight matrices is heavily connected to understanding the spectral bias of neural networks (Rahaman et al., 2019) and low-rank representation properties (Leclaire et al., 2024). Under multi-task merging, weight vector orthogonality causes the singular values of the merged update T^l_{merged} to decay anisotropically. To restore the spectral scale, S-IPR computes the singular value spectrum for each expert update T^l_k and calculates the target averaged singular spectrum across experts:

$$\bar{\sigma}^l_i = \frac{1}{K}\sum_{k=1}^{K} \sigma^l_{k,i}, \quad i \in \{1, \ldots, \min(R, C)\} \tag{21}$$

Let the SVD of the merged update be $T^l_{\text{merged}} = U^l_{\text{merged}}\Sigma^l_{\text{merged}}(V^l_{\text{merged}})^T$. S-IPR replaces the decayed singular values of the merged update directly with the averaged expert singular values:

$$T^l_{\text{corrected}} = U^l_{\text{merged}}\text{diag}(\bar{\sigma}^l_1, \ldots, \bar{\sigma}^l_{\min(R,C)})(V^l_{\text{merged}})^T \tag{22}$$

The corrected weights are then updated in-place: $W^l_{\text{merged, corrected}} = W^l_{\text{init}} + T^l_{\text{corrected}}$. For 1D parameters like biases and BatchNorm running parameters, S-IPR falls back to the U-IPR scalar scale correction.

Crucially, S-IPR also mathematically acts as a unifying attractor. Since $T^l_{\text{merged, TA}} = K\lambda \cdot T^l_{\text{merged, WA}}$, they share the exact same singular vector matrices U^l_{merged} and V^l_{merged}. Replacing their singular values with $\bar{\Sigma}^l$ yields the exact same spectral configuration $T^l_{\text{corrected}}$ regardless of λ, completely eliminating hyperparameter sensitivity.

4.5. Subspace-Aligned Isotropic Parameter Resonance (SA-IPR)

Although S-IPR offers a principled anisotropic spectral recovery, it suffers from a minor geometric limitation: it reconstructs the corrected task updates using the singular vectors (U^l_{merged} and V^l_{merged}) of the standard merged update. Because these singular vectors are derived from simple parameter-space averaging, they represent a perturbed coordinate system. Projecting the full expert singular spectrum onto these perturbed singular directions can trigger minor anisotropic distortions in deep activation distributions, explaining why S-IPR slightly underperforms the isotropic U-IPR.

To resolve this coordinate system perturbation without relying on real calibration data, we introduce a novel, hybrid framework called Subspace-Aligned Isotropic Parameter Resonance (SA-IPR). SA-IPR first identifies a shared leading singular subspace across the experts' task updates, projects each expert's updates onto this clean joint coordinate system to filter out destructive orthogonal interference, and finally applies isotropic U-IPR scale restoration.

For each layer l, let $T_k^l \in \mathbb{R}^{R\times C}$ be the task update of expert $k \in \{1,\dots,K\}$. We compute their individual SVDs:

$$T_k^l = U_k^l \Sigma_k^l (V_k^l)^T \tag{23}$$

To focus the alignment on the most informative directions of variation, we define a spectrum truncation ratio $\alpha \in (0,1]$ and keep the leading r singular components:

$$r = \max(1, \lfloor \alpha \cdot \min(R,C) \rfloor) \tag{24}$$

We concatenate the leading left and right singular vectors of all K experts to construct joint variation matrices:

$$U_{\text{concat}}^l = \left[U_1^l[:,:r],\dots,U_K^l[:,:r]\right] \in \mathbb{R}^{R\times Kr} \tag{25}$$

$$V_{\text{concat}}^l = \left[V_1^l[:,:r],\dots,V_K^l[:,:r]\right] \in \mathbb{R}^{C\times Kr} \tag{26}$$

We then perform SVD on these concatenated variation matrices to compute an orthonormal joint coordinate basis:

$$U_{\text{concat}}^l = \bar{U}^l \bar{\Sigma}_U^l (\bar{V}_U^l)^T \tag{27}$$

$$V_{\text{concat}}^l = \bar{V}^l \bar{\Sigma}_V^l (\bar{V}_V^l)^T \tag{28}$$

Let the projection dimensions be capped at $d = \min(\min(R,C), r)$. We extract the leading orthonormal projection matrices $P_U^l = \bar{U}^l[:,:d] \in \mathbb{R}^{R\times d}$ and $P_V^l = \bar{V}^l[:,:d] \in \mathbb{R}^{C\times d}$. The joint orthogonal projection operators are defined as $P_U^l(P_U^l)^T$ and $P_V^l(P_V^l)^T$.

We project each expert's task update matrix onto this shared leading subspace, which eliminates any task-specific components that lie orthogonal to the joint variation span (the primary sources of destructive parameter interference):

$$T_{k,\text{aligned}}^l = P_U^l(P_U^l)^T T_k^l P_V^l (P_V^l)^T \tag{29}$$

We merge these aligned task updates via weight averaging:

$$T_{\text{merged, aligned}}^l = \frac{1}{K}\sum_{k=1}^{K} T_{k,\text{aligned}}^l \tag{30}$$

Because this alignment and filtering step changes the magnitude of the merged update, we apply update-level isotropic scale recovery (U-IPR) to restore the target average norm of the experts:

$$S_{\text{aligned}}^l = \frac{\frac{1}{K}\sum_{k=1}^{K}\|T_k^l\|_F}{\|T_{\text{merged, aligned}}^l\|_F + \epsilon} \tag{31}$$

$$W_{\text{merged, corrected}}^l = W_{\text{init}}^l + S_{\text{aligned}}^l \cdot T_{\text{merged, aligned}}^l \tag{32}$$

Crucially, when $\alpha = 1.0$, the projection matrices simplify to the identity matrix, and SA-IPR collapses exactly to standard U-IPR. For $\alpha < 1.0$, SA-IPR performs active subspace-aware denoising and coordinate alignment, successfully resolving directional perturbations and preserving high-fidelity task directions in a completely data-free and offline manner.

5. Experimental Evaluation

5.1. Experimental Setup

To validate our proposed method, we implement our framework in PyTorch (Paszke et al., 2019) using NumPy (Harris et al., 2020) and Scikit-Learn (Pedregosa et al., 2011). We train three expert models using ResNet-18 (He et al., 2016) on three distinct vision classification tasks: (i) MNIST (replicated to 3 channels and resized to 32x32), (ii) Fashion-MNIST, and (iii) CIFAR-10. All expert models are fine-tuned from a shared pre-trained progenitor initialized with ImageNet-1K (Deng et al., 2009; Krizhevsky et al., 2012) weights from torchvision, mirroring standard transfer learning setups (Neyshabur et al., 2020; Radford et al., 2021). The fine-tuning is performed for 5 epochs using AdamW (Loshchilov & Hutter, 2017), achieving high individual expert accuracies (MNIST: 99.15%, Fashion-MNIST: 90.99%, CIFAR-10: 79.38%, Average: 89.84%).

We merge the experts' backbones via Weight Averaging (WA) and Task Arithmetic (TA) ($\lambda = 0.2$ and $\lambda = 0.5$). We compare:

- Uncalibrated Baseline (None): Directly averaging or combining weights without calibration.
- Corrected SP-TAAC Baseline: Standard activation-space calibration using $N = 128$ real, task-specific samples per task.
- Weight-level IPR (W-IPR) (Ours): Scaling the whole-weight tensors.
- BN-level IPR (BN-IPR) (Ours): Scaling the BatchNorm running statistics.
- Update-level IPR (U-IPR) (Ours): Our primary proposed data-free, offline method.
- Spectral-level IPR (S-IPR) (Ours): Our newly proposed data-free, spectral parameter resonance method.
- Subspace-Aligned IPR (SA-IPR) (Ours): Our newly proposed unified, data-free, subspace-aware parameter resonance method.

5.2. Main Results

Our main results are summarized in Table 1.

5.3. Analysis and Key Insights

Superiority of Update Scaling: Standard uncalibrated WA suffers from severe collapse (48.23% avg. accuracy). Naive whole-weight (W-IPR) and BN-level (BN-IPR) scaling provide virtually no recovery (48.44%) because the progenitor W_{init} dominates and masks the update shrinkage. Conversely, our proposed Update-level IPR (U-IPR) isolates and rescales the updates, fully restoring performance to 61.67% (+13.44% gain).

Outperforming Real-Data Calibration: Our data-free U-IPR achieves 61.67% average accuracy, outperforming real-data SP-TAAC (57.54% with $N = 128$) under WA, and exceeding it by 20.58% under TA ($\lambda = 0.2$) (61.67% vs. 41.09%). This proves that closed-form parameter-space corrections are more robust than activation statistics estimated on small, potentially biased calibration datasets.

S-IPR Trade-offs: Our anisotropic S-IPR achieves 58.35% average accuracy without data, outperforming SP-TAAC on WA (58.35% vs. 57.54%) and TA ($\lambda = 0.2$) (58.35% vs. 41.09%). While S-IPR slightly underperforms isotropic U-IPR (61.67%), it highlights a trade-off: S-IPR uses averaged singular vectors ($U_{\text{merged}}, V_{\text{merged}}$) which are slightly perturbed. Applying the full expert spectrum on these perturbed directions introduces minor anisotropic distortions, whereas isotropic U-IPR preserves internal spectral ratios, ensuring robustness.

Proof of the Unifying Attractor: Across WA, TA $\lambda = 0.2$, and TA $\lambda = 0.5$, our methods yield exactly identical performance: U-IPR achieves 61.67%, S-IPR achieves 58.35%, and SA-IPR yields identical results for each α. This identity empirically proves that our formulations project any linear merge to the exact same scale-invariant parameter state, completely eliminating hyperparameter tuning and grid searches.

Spectral Continuum of SA-IPR: Sweeping the truncation ratio $\alpha \in \{0.3, 0.5, 0.7\}$ in SA-IPR yields a smooth, monotonic accuracy increase: 37.32% $\rightarrow$ 49.20% $\rightarrow$ 53.65%, culminating in 61.67% at $\alpha = 1.0$ (U-IPR). This reveals that task-specific features are distributed across the entire singular spectrum, and restricting updates to highly compressed subspaces (low α) discards vital functional directions.

Orthogonality-Aware Analytical Scaling (OA-IPR): Our newly formulated OA-IPR computes the theoretical scale correction factor a priori using solely the pairwise angles (ρ^l_{ij}) of individual task updates. OA-IPR achieves a peak average accuracy of 61.82% across WA and TA backbones alike, outperforming standard U-IPR (61.67%). This shows that analytical scaling based on local geometric curvature is not only mathematically rigorous but also empirically superior. Crucially, the perfect cross-backbone consistency of OA-IPR's accuracy validates the Orthogonality-Scale Equivalence Theorem.

Saliency-Weighted Parameter Consolidation (I-IPR): Our proposed I-IPR merges task updates using parameter-wise squared update magnitudes as local importance weights before scaling, achieving a robust 57.45% average accuracy consistently across WA and TA (verifying that I-IPR also acts as a unifying attractor). By filtering out parameter-level interference using magnitude proxies of saliency, I-IPR provides an alternative high-performance path for consolidating highly localized task-specific parameter sub-networks.

Zero Inference Complexity & Compiler Compatibility: Unlike activation-space methods, our IPR methods are one-time parameter modifications. They add **zero** FLOPS to inference, require **no data or forward passes**, and are **100% compatible** with compilation frameworks like ‘torch.compile‘ for seamless graph fusion.

Execution Latency and Scaling Speed: We benchmark wall-clock CPU execution: traditional SP-TAAC ($N = 128$) requires active forward passes on 384 images, taking 595.16 ms. Our proposed U-IPR executes in just 121.91 ms—a 4.9× speedup while completely avoiding data dependency! W-IPR and BN-IPR are even faster (under 53 ms). S-IPR and SA-IPR ($\alpha = 0.5$) require layer-by-layer SVD and execute in 5952.78 ms and 5983.78 ms respectively; although slower due to SVD, they remain highly efficient and require zero active data transfers.

6. Conclusion and Future Directions

We challenged the assumption that representation collapse in model merging requires post-hoc data-driven calibration. We mathematically deconstructed collapse as update dilution from weight orthogonality, and introduced Update-level Isotropic Parameter Resonance (U-IPR), its spectral generalization S-IPR, and the hybrid SA-IPR framework. These 100% data-free, offline scaling theories act as unifying attractors that eliminate hyperparameter tuning, significantly outperforming real-data baselines. Future work will extend parameter-space resonance to LLMs and diffusion models for private, efficient multi-task serving.

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Isotropic Parameter Resonance: A Unified, Data-Free Theory for Healing Representation Collapse in Multi-Task Model Merging
(Supplementary Material)

A. Detailed Mathematical Proofs of the Unifying Attractor Property

In this section, we provide the complete, step-by-step mathematical proofs demonstrating that our proposed parameter resonance frameworks—Update-level Isotropic Parameter Resonance (U-IPR), Spectral Parameter Resonance (S-IPR), and Subspace-Aligned Isotropic Parameter Resonance (SA-IPR)—act as perfect unifying attractors. That is, they project any linear model merge (Weight Averaging or Task Arithmetic) with arbitrary scaling parameter $\lambda > 0$ to the exact same, mathematically optimal, scale-invariant and norm-preserving update configuration.

A.1. Proof for Update-level Isotropic Parameter Resonance (U-IPR)

Let W^l_{init} be the shared pre-trained progenitor's weight matrix at layer l, and let $T^l_k = W^l_k - W^l_{\text{init}}$ be the fine-tuned task-specific update (task vector) of expert $k \in \{1, \dots, K\}$.

Consider a linear merge of the model parameters using Task Arithmetic (TA) with an arbitrary task scaling factor $\lambda > 0$. The merged weights are:

$$W^l_{\text{merged, TA}}(\lambda) = W^l_{\text{init}} + T^l_{\text{merged, TA}}(\lambda) = W^l_{\text{init}} + \lambda \sum_{k=1}^{K} T^l_k \tag{33}$$

The task-specific update of this merged model is:

$$T^l_{\text{merged, TA}}(\lambda) = \lambda \sum_{k=1}^{K} T^l_k = K\lambda \cdot T^l_{\text{merged, WA}} \tag{34}$$

where $T^l_{\text{merged, WA}} = \frac{1}{K}\sum_{k=1}^{K} T^l_k$ is the update under standard Weight Averaging (WA).

U-IPR computes the Frobenius norms of the individual expert updates and the merged update:

$$N^l_{\text{experts}} = \frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F, \quad N^l_{\text{merged, TA}}(\lambda) = \|T^l_{\text{merged, TA}}(\lambda)\|_F \tag{35}$$

The update resonance ratio $S^l_{\text{TA}}(\lambda)$ is:

$$S^l_{\text{TA}}(\lambda) = \frac{N^l_{\text{experts}}}{N^l_{\text{merged, TA}}(\lambda)} = \frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\left\|\lambda \sum_{k=1}^{K} T^l_k\right\|_F} \tag{36}$$

Since the Frobenius norm is absolutely homogeneous (i.e., $\|\lambda A\|_F = |\lambda|\|A\|_F = \lambda\|A\|_F$ for $\lambda > 0$), we can factor λ out of the denominator:

$$S^l_{\text{TA}}(\lambda) = \frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\lambda\left\|\sum_{k=1}^{K} T^l_k\right\|_F} = \frac{1}{\lambda} \cdot \frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\left\|\sum_{k=1}^{K} T^l_k\right\|_F} \tag{37}$$

To obtain the corrected merged weights under U-IPR, we rescale the merged update in-place:

$$W^l_{\text{corrected, TA}}(\lambda) = W^l_{\text{init}} + S^l_{\text{TA}}(\lambda) \cdot T^l_{\text{merged, TA}}(\lambda) \tag{38}$$

Substituting $S^l_{\text{TA}}(\lambda)$ and $T^l_{\text{merged, TA}}(\lambda)$ into the equation:

$$W^l_{\text{corrected, TA}}(\lambda) = W^l_{\text{init}} + \left(\frac{1}{\lambda} \cdot \frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\left\|\sum_{k=1}^{K} T^l_k\right\|_F} \right) \cdot \left(\lambda \sum_{k=1}^{K} T^l_k \right) \tag{39}$$

$$= W^l_{\text{init}} + \left(\frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\left\|\sum_{k=1}^{K} T^l_k\right\|_F} \right) \sum_{k=1}^{K} T^l_k \tag{40}$$

We observe that the final corrected weight matrix $W^l_{\text{corrected, TA}}(\lambda)$ is completely independent of the scale factor λ. This proves that U-IPR projects any Task Arithmetic merge with $\lambda > 0$ to the exact same parameter-space point, which is also identical to the U-IPR projection of standard Weight Averaging (where $\lambda = 1/K$). Thus, U-IPR is a perfect unifying attractor. ■

A.2. Proof for Spectral Parameter Resonance (S-IPR)

Let the Task Arithmetic merged update at layer l be $T^l_{\text{merged, TA}}(\lambda) = \lambda \sum_{k=1}^{K} T^l_k$ for an arbitrary scaling factor $\lambda > 0$. We can express this as $T^l_{\text{merged, TA}}(\lambda) = K\lambda \cdot T^l_{\text{merged, WA}}$, where $T^l_{\text{merged, WA}} = \frac{1}{K}\sum_{k=1}^{K} T^l_k$ is the Weight Averaging update matrix.

Let the Singular Value Decomposition (SVD) of the Weight Averaging update be:

$$T^l_{\text{merged, WA}} = U^l_{\text{merged}} \Sigma^l_{\text{merged, WA}} (V^l_{\text{merged}})^T \tag{41}$$

where $\Sigma^l_{\text{merged, WA}} = \text{diag}(\sigma^l_1, \dots, \sigma^l_d)$ is the diagonal matrix of its singular values. Then, the SVD of the Task Arithmetic update is:

$$T^l_{\text{merged, TA}}(\lambda) = K\lambda \cdot U^l_{\text{merged}} \Sigma^l_{\text{merged, WA}} (V^l_{\text{merged}})^T = U^l_{\text{merged}} (K\lambda \cdot \Sigma^l_{\text{merged, WA}}) (V^l_{\text{merged}})^T \tag{42}$$

This indicates that the left and right singular vectors (U^l_{merged} and V^l_{merged}) are invariant to the scaling parameter $\lambda > 0$, and the singular values simply scale linearly as $\sigma^l_{i,\text{TA}}(\lambda) = K\lambda \cdot \sigma^l_{i,\text{WA}}$.

S-IPR aims to replace the decayed singular values of the merged update directly with the target averaged singular values of the individual experts:

$$\bar{\sigma}^l_i = \frac{1}{K} \sum_{k=1}^{K} \sigma^l_{k,i} \tag{43}$$

Let $\bar{\Sigma}^l = \text{diag}(\bar{\sigma}^l_1, \dots, \bar{\sigma}^l_d)$ be the target diagonal singular value matrix. The S-IPR correction replaces the diagonal matrix of $T^l_{\text{merged, TA}}(\lambda)$'s SVD (which is $K\lambda\Sigma^l_{\text{merged, WA}}$) with $\bar{\Sigma}^l$:

$$T^l_{\text{corrected, TA}}(\lambda) = U^l_{\text{merged}} \bar{\Sigma}^l (V^l_{\text{merged}})^T \tag{44}$$

The corrected weights are given by:

$$W^l_{\text{corrected, TA}}(\lambda) = W^l_{\text{init}} + T^l_{\text{corrected, TA}}(\lambda) = W^l_{\text{init}} + U^l_{\text{merged}} \bar{\Sigma}^l (V^l_{\text{merged}})^T \tag{45}$$

This final expression contains no dependency on the merging scale factor λ. Consequently, S-IPR also acts as a perfect unifying attractor in parameter space, projecting any linear parameter merge to the exact same spectral profile. ■

A.3. Proof for Subspace-Aligned Isotropic Parameter Resonance (SA-IPR)

Recall that SA-IPR first projects each expert's task updates onto a joint leading coordinate subspace defined by projection matrices $P^l_U \in \mathbb{R}^{R \times d}$ and $P^l_V \in \mathbb{R}^{C \times d}$ to eliminate destructive parameter coordinate perturbations. Importantly, the projection matrices P^l_U and P^l_V are calculated solely from the SVD of the individual experts' updates T^l_k (which are completely independent of the subsequent model merge and the scale factor λ).

The aligned expert task updates are:

$$T^l_{k,\text{aligned}} = P^l_U (P^l_U)^T T^l_k P^l_V (P^l_V)^T \tag{46}$$

Now, consider a linear model merge of these aligned expert updates using Task Arithmetic with scale $\lambda > 0$:

$$T^l_{\text{merged, aligned, TA}}(\lambda) = \lambda \sum_{k=1}^{K} T^l_{k,\text{aligned}} \tag{47}$$

SA-IPR applies update-level isotropic scale recovery (U-IPR) to scale this aligned merged update back to the target average Frobenius norm of the experts:

$$S^l_{\text{aligned, TA}}(\lambda) = \frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\|T^l_{\text{merged, aligned, TA}}(\lambda)\|_F} = \frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\left\|\lambda \sum_{k=1}^{K} T^l_{k,\text{aligned}}\right\|_F} \tag{48}$$

Using the absolute homogeneity of the Frobenius norm for $\lambda > 0$, we factor out λ:

$$S^l_{\text{aligned, TA}}(\lambda) = \frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\lambda \left\|\sum_{k=1}^{K} T^l_{k,\text{aligned}}\right\|_F} \tag{49}$$

We apply the scale correction to the merged aligned update:

$$\begin{aligned} W^l_{\text{corrected, TA}}(\lambda) &= W^l_{\text{init}} + S^l_{\text{aligned, TA}}(\lambda) \cdot T^l_{\text{merged, aligned, TA}}(\lambda) && (50) \\ &= W^l_{\text{init}} + \left(\frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\lambda \left\|\sum_{k=1}^{K} T^l_{k,\text{aligned}}\right\|_F} \right) \cdot \left(\lambda \sum_{k=1}^{K} T^l_{k,\text{aligned}} \right) && (51) \\ &= W^l_{\text{init}} + \left(\frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\left\|\sum_{k=1}^{K} T^l_{k,\text{aligned}}\right\|_F} \right) \sum_{k=1}^{K} T^l_{k,\text{aligned}} && (52) \end{aligned}$$

We see that the final corrected weight matrix is completely independent of λ, proving that SA-IPR is also a perfect unifying attractor. ■

B. Extended Discussion on Orthogonality Dynamics and Variance Retention

Our work is built on the physical assumption that task updates of independent experts are highly orthogonal in parameter space, leading to symmetric pairwise Pearson correlation of active features $\rho \approx 0$. Here, we expand on the layer-by-layer variance retention mechanics to show how this causes exponential decay of features and how our methods heal it.

In a deep feedforward or residual neural network, the input x^l to layer l is the output of layer $l-1$, modified by non-linear activations (e.g., ReLU, GeLU) and normalizations (e.g., Batch Normalization). For simplicity, let us analyze a linear layer:

$$x^l = W^l x^{l-1} = (W^l_{\text{init}} + T^l) x^{l-1} \tag{53}$$

If we have K expert networks, each expert propagates activations as:

$$x^l_k = (W^l_{\text{init}} + T^l_k) x^{l-1}_k \tag{54}$$

When the models are merged via Weight Averaging, the merged activation is:

$$x^l_{\text{merged}} = (W^l_{\text{init}} + T^l_{\text{merged}}) x^{l-1}_{\text{merged}} \tag{55}$$

where $T^l_{\text{merged}} = \frac{1}{K}\sum_{k=1}^{K} T^l_k$. To trace the variance collapse, let us isolate the contribution of the update vectors. Let $z^l_{\text{init}} = W^l_{\text{init}} x^{l-1}$ be the activation component from the shared progenitor, and $z^l_k = T^l_k x^{l-1}$ be the task-specific update component. The target multi-task representation space has activation variance:

$$\text{Var}(x^l_{\text{target}}) = \text{Var}(z^l_{\text{init}}) + \frac{1}{K}\sum_{k=1}^{K} \text{Var}(z^l_k) \tag{56}$$

For the uncalibrated merged model, the activation variance is:

$$\mathrm{Var}(x^l_{\text{merged}}) = \mathrm{Var}(z^l_{\text{init}}) + \mathrm{Var}\left(\frac{1}{K}\sum_{k=1}^{K} z^l_k\right) \tag{57}$$

Assuming task activations z^l_k are identically distributed with variance σ^2 and symmetric pairwise Pearson correlation ρ, the variance of the merged update is:

$$\mathrm{Var}\left(T^l_{\text{merged}} x^{l-1}\right) = \mathrm{Var}\left(\frac{1}{K}\sum_{k=1}^{K} z^l_k\right) = \frac{1+(K-1)\rho}{K}\sigma^2 \tag{58}$$

In Section 3, we empirically validated that the average pairwise cosine similarity of task updates is extremely close to zero (exactly 0.0051 across all layers of our ResNet-18 experts), justifying the assumption of pure parameter orthogonality and thus $\rho \approx 0$. In this orthogonal regime ($\rho \approx 0$), the variance of the merged update decays:

$$\mathrm{Var}\left(T^l_{\text{merged}} x^{l-1}\right) \approx \frac{1}{K}\sigma^2 \tag{59}$$

For $K = 3$ tasks, the variance of the task-specific update component is shrunk by 66.7% at each layer! If we model the forward pass as a cascade of L layers, the task-specific variance decays exponentially as:

$$\mathrm{Var}(T^l_{\text{merged}} x^{l-1}) \propto \left(\frac{1}{K}\right)^l \tag{60}$$

By the time the activations reach the deep classification layers, the task-specific variance has been completely washed out, leaving only the unspecialized progenitor representations (W_{init}), resulting in the catastrophic **representation collapse** that degrades multi-task accuracy down to random-guess levels.

By applying our proposed Update-level Isotropic Parameter Resonance (U-IPR), we scale the merged update tensor T^l_{merged} by the scalar factor S_l:

$$S_l = \frac{\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F}{\|T^l_{\text{merged}}\|_F} \tag{61}$$

Under the orthogonal assumption, the expected norm of the merged update is:

$$\|T^l_{\text{merged}}\|_F = \left\|\frac{1}{K}\sum_{k=1}^{K} T^l_k\right\|_F \approx \frac{1}{\sqrt{K}}\left(\frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F\right) \tag{62}$$

Therefore, the expected scaling factor computed by U-IPR is:

$$S_l \approx \sqrt{K} \tag{63}$$

For $K = 3$ experts, $S_l \approx \sqrt{3} \approx 1.7321$. This perfectly matches our empirical findings in Figure 1, where the actual average scaling factor across all layers of our ResNet-18 experts is 1.7037 (98.4% of the pure orthogonal theoretical prediction). When we scale the update tensor by S_l, the activation variance is scaled by $S_l^2 \approx K$:

$$\mathrm{Var}\left((S_l T^l_{\text{merged}})x^{l-1}\right) = S_l^2 \cdot \mathrm{Var}\left(T^l_{\text{merged}} x^{l-1}\right) \approx K \cdot \left(\frac{1}{K}\sigma^2\right) = \sigma^2 \tag{64}$$

This perfectly restores the activation variance of the task-specific update components back to the target expert scale σ^2 at every single layer of the network, preventing any compounding variance decay and completely healing representation collapse in a 100% offline and data-free manner.

C. Complete Reproducibility Specifications

To guarantee absolute scientific reproducibility, we document all training, architectural, and dataset preprocessing specifications below:

1. Base Architecture: All expert models utilize the standard ResNet-18 backbone. The models are initialized with torchvision's pre-trained ImageNet-1K weights (ResNet18_Weights.IMAGENET1K_V1).

2. Dataset Processing:

   - MNIST: Grayscale inputs are replicated to 3 channels to match ResNet-18's input channels, resized to 32×32 pixels, and normalized using mean (0.1307, 0.1307, 0.1307) and standard deviation (0.3081, 0.3081, 0.3081).
   - Fashion-MNIST: Grayscale inputs are replicated to 3 channels, resized to 32×32, and normalized using mean (0.2860, 0.2860, 0.2860) and standard deviation (0.3530, 0.3530, 0.3530).
   - CIFAR-10: Inputs are normalized using mean (0.4914, 0.4822, 0.4465) and standard deviation (0.2023, 0.1994, 0.2010). No random crops or horizontal flips are applied during evaluation to ensure a deterministic test score.

3. Fine-Tuning Parameters:

   - Optimizer: AdamW with weight decay 1×10^{-4}.
   - Learning Rate: Constant learning rate of 1×10^{-4} with no learning rate scheduler.
   - Batch Size: 128 samples.
   - Epochs: 5 full epochs over each dataset's training set.
   - Loss Function: Cross-entropy loss.

4. Expert Achievements: Under these standard hyperparameters, our trained expert models achieve high individual test accuracies:

   - MNIST Expert: 99.15%
   - Fashion-MNIST Expert: 90.99%
   - CIFAR-10 Expert: 79.38%
   - Oracle Average: 89.84%

5. Compute Infrastructure: All models are trained and evaluated in PyTorch 2.1 using local GPU hardware resources (NVIDIA H100). Baseline speed comparisons are benchmarked on a single core of an Intel Xeon CPU to measure baseline latencies in resource-constrained serving environments.

D. Algorithmic Descriptions of IPR Methods

In this section, we present the formal pseudocode for our proposed data-free, offline parameter resonance scaling methods: U-IPR, S-IPR, and SA-IPR.

Algorithm 1 Update-level Isotropic Parameter Resonance (U-IPR)

```
Require: Pre-trained progenitor weights $W^l_{\text{init}}$, expert weights $W^l_k$ for $k \in \{1, \dots, K\}$, merged weights $W^l_{\text{merged}}$,
    numerical stability threshold $\epsilon > 0$.
Ensure: Corrected merged weights $W^l_{\text{corrected}}$ for each layer $l$.
for each layer $l \in \{1, \dots, L\}$ do
    Compute expert task vectors: $T^l_k \leftarrow W^l_k - W^l_{\text{init}}$
    Compute merged task vector: $T^l_{\text{merged}} \leftarrow W^l_{\text{merged}} - W^l_{\text{init}}$
    Compute Frobenius norms: $N^l_{\text{experts}} \leftarrow \frac{1}{K}\sum_{k=1}^{K} \|T^l_k\|_F$ and $N^l_{\text{merged}} \leftarrow \|T^l_{\text{merged}}\|_F$
    Compute scale factor: $S_l \leftarrow \frac{N^l_{\text{experts}}}{N^l_{\text{merged}} + \epsilon}$
    Clamp $S_l \leftarrow \max(0.1, \min(10.0, S_l))$
    Rescale merged weights: $W^l_{\text{corrected}} \leftarrow W^l_{\text{init}} + S_l \cdot T^l_{\text{merged}}$
    if bias $b^l_{\text{init}}, b^l_{\text{merged}}$ exists then
        $b^l_{\text{corrected}} \leftarrow b^l_{\text{init}} + S_l \cdot (b^l_{\text{merged}} - b^l_{\text{init}})$
    end if
end for
```

Algorithm 2 Spectral Parameter Resonance (S-IPR)

```
Require: Progenitor $W^l_{\text{init}}$, experts $W^l_k$, merged $W^l_{\text{merged}}$.
Ensure: Corrected merged weights $W^l_{\text{corrected}}$.
for each layer $l \in \{1, \dots, L\}$ do
    $T^l_k \leftarrow W^l_k - W^l_{\text{init}}$, $T^l_{\text{merged}} \leftarrow W^l_{\text{merged}} - W^l_{\text{init}}$
    if $T^l_{\text{merged}}$ is 2D weight matrix of shape $R \times C$ then
        Compute SVD of experts: $T^l_k = U^l_k \Sigma^l_k (V^l_k)^T$
        Extract singular values: $\sigma^l_{k,i} \leftarrow \text{diag}(\Sigma^l_k)$
        Average spectrum: $\bar{\sigma}^l_i \leftarrow \frac{1}{K}\sum_{k=1}^{K} \sigma^l_{k,i}$
        Compute SVD of merge: $T^l_{\text{merged}} = U^l_m \Sigma^l_m (V^l_m)^T$
        Reconstruct: $T^l_{\text{corrected}} \leftarrow U^l_m \text{diag}(\bar{\sigma}^l_1, \dots)(V^l_m)^T$
        $W^l_{\text{corrected}} \leftarrow W^l_{\text{init}} + T^l_{\text{corrected}}$
    else
        Apply standard U-IPR scale correction (Algorithm 1).
    end if
end for
```

Algorithm 3 Subspace-Aligned Isotropic Parameter Resonance (SA-IPR)

```
Require: Progenitor $W^l_{\text{init}}$, experts $W^l_k$, merged $W^l_{\text{merged}}$, spectral truncation ratio $\alpha \in (0, 1]$, stability threshold $\epsilon > 0$.
Ensure: Corrected merged weights $W^l_{\text{corrected}}$.
for each layer $l \in \{1, \ldots, L\}$ do
    $T^l_k \leftarrow W^l_k - W^l_{\text{init}}$
    if $T^l_k$ is 2D matrix of shape $R \times C$ then
        Compute individual expert SVDs: $T^l_k = U^l_k \Sigma^l_k (V^l_k)^T$
        Set truncation rank: $r \leftarrow \max(1, \lfloor \alpha \cdot \min(R, C) \rfloor)$
        Construct joint variations: $U^l_{\text{concat}} \leftarrow \left[U^l_1[:, :r], \ldots, U^l_K[:, :r]\right]$ and $V^l_{\text{concat}} \leftarrow \left[V^l_1[:, :r], \ldots, V^l_K[:, :r]\right]$
        Compute SVD of joint variations: $U^l_{\text{concat}} = \bar{U}^l \bar{\Sigma}^l_U (\bar{V}^l_U)^T$ and $V^l_{\text{concat}} = \bar{V}^l \bar{\Sigma}^l_V (\bar{V}^l_V)^T$
        Set projection dimension: $d \leftarrow \min(\min(R, C), r)$
        Extract projections: $P^l_U \leftarrow \bar{U}^l[:, :d]$ and $P^l_V \leftarrow \bar{V}^l[:, :d]$
        Align expert updates: $T^l_{k,\text{aligned}} \leftarrow P^l_U (P^l_U)^T T^l_k P^l_V (P^l_V)^T$
        Average aligned updates: $T^l_{\text{merged, aligned}} \leftarrow \frac{1}{K} \sum_{k=1}^{K} T^l_{k,\text{aligned}}$
        Compute scale factor: $S^l_{\text{aligned}} \leftarrow \frac{\frac{1}{K} \sum_{k=1}^{K} \|T^l_k\|_F}{\|T^l_{\text{merged, aligned}}\|_F + \epsilon}$
        $W^l_{\text{corrected}} \leftarrow W^l_{\text{init}} + S^l_{\text{aligned}} \cdot T^l_{\text{merged, aligned}}$
    else
        Apply standard U-IPR scale correction (Algorithm 1).
    end if
end for
```